

Q2 2024 Earnings Call

Company Participants

- Brice Hill, Senior Vice President, Chief Financial Officer and Global Information Services
- Gary E. Dickerson, President and Chief Executive Officer
- Michael Sullivan, Corporate Vice President

Other Participants

- Atif Malik, Citi
- C.J. Muse, Cantor Fitzgerald
- Chris Caso, Wolfe Research
- Harlan Sur, JP Morgan
- Joe Quatrochi, Wells Fargo
- Krish Sankar, TD Cowen
- Mark Lipacis, Evercore ISI
- Mehdi Hosseini, Susquehanna Financial Group
- Srinii Pajjuri, Raymond James
- Stacy Rasgon, Bernstein
- Timothy Arcuri, UBS Securities
- Toshiya Hari, Goldman Sachs and Company
- Vivek Arya, Bank of America

Presentation

Operator

Welcome to the Applied Materials' Earnings Conference Call. During the presentation, all participants will be in a listen-only mode. Afterwards, you will be invited to participate in a question-and-answer session.

I would now like to turn the conference over to Michael Sullivan, Corporate Vice President. Please go ahead, sir.

Michael Sullivan {**BIO 16341622 <GO>**}

Good afternoon, everyone, and thank you for joining Applied's second quarter of fiscal 2024 earnings call.

Joining me are Gary Dickerson, our President & CEO; and Brice Hill, our Chief Financial Officer.

Before we begin, I'd like to remind you that today's call contains forward-looking statements, which are subject to risks and uncertainties that could cause our actual results to differ. Information concerning the risks and uncertainties is contained in Applied's most recent Form 10-Q filing with the SEC. Today's call also includes non-GAAP financial measures. Reconciliations to GAAP measures are found in today's earnings press release and in our quarterly earnings materials, which are available on our website at ir.appliedmaterials.com.

Before we begin, I have a calendar announcement. On Tuesday morning, July 9th, from 7:30 to 9:00 A.M., Applied will host a technology breakfast event at SEMICON West in San Francisco. Joining us will be Dr. Prabu Raja, President of Applied's Semiconductor Products Group; along with Mark Fuselier, who is Senior Vice President of Technology and Product Engineering at AMD. After Mark shares AMD's AI computing technology vision, Applied's experts will share our advanced materials engineering roadmap for making future AI chips. We'll outline device architecture inflections in logic, including transistors, frontside wiring and backside power, along with DRAM, high-bandwidth memory, and other forms of advanced packaging. We hope you'll join us.

And with that introduction, I'd like to turn the call over to Gary Dickerson.

Gary E. Dickerson {**BIO 2135669 <GO>**}

Thank you, Mike.

With second quarter revenue and earnings toward the high-end of our guided range, Applied Materials continues to deliver strong performance in 2024, and we're in a great position to benefit from secular growth trends over the longer-term. Semiconductors are the foundation of huge technology trends reshaping the global economy. These trends are driving demand for more chip manufacturing capacity, as well as better chips with higher performance and improved energy efficiency.

Key inflections that underpin the semiconductor roadmap are enabled by Applied Materials and will support our ongoing outperformance as next-generation chip technologies move into high-volume production.

In addition, the complexity of implementing the industry's roadmap and bringing new semiconductor technologies to market is driving earlier, deeper, and broader collaboration with customers, as well as supporting double-digit growth for our service business.

In my prepared remarks today, I'll provide some examples of how Applied's innovations are helping to enable multi-trillion-dollar technology inflections, including AI. I'll explain how this translates to our performance in the near-term and our future growth potential. I'll also describe how we're working with our customers and partners to accelerate major semiconductor inflections all the way from research and development to high-volume manufacturing, and by doing this, how we are capturing more value in our service business.

I'll start with a big picture perspective. Tectonic shifts in technology, including AI, IoT and automation, electric and autonomous vehicles, and clean energy will transform virtually every area of economy over the next several decades. And they all have one thing in common. They are built upon semiconductors. As these new technologies are deployed, they are driving growth and innovation across the semiconductor ecosystem. In terms of impact and scale, I believe AI will be the biggest technology inflection of our lifetimes, and at the heart of AI are some of the world's most sophisticated chips.

In simple terms, the advanced chips that power AI data centers are enabled by four key semiconductor technologies: leading-edge logic; compute memory, or high-performance DRAM; DRAM stacking technology, referred to as high-bandwidth memory, or HBM; and advanced packaging to connect the logic and memory chips together and create a system in a package.

Applied has process technology leadership in all four of these areas, and we have made significant investments in next-generation solutions to make possible the key device architecture inflections that are essential for our customers' future roadmaps.

In advanced logic, Applied has longstanding leadership in the materials engineering processes for both transistors and interconnects. The first nodes that use Gate-All-Around transistors are now transitioning to high-volume manufacturing. These new transistor process flows are considerably more complex, and the shift from FinFET to Gate-All-Around grows Applied's available market for the transistor module from around \$6 billion to approximately \$7 billion for every 100,000 wafer starts per month of capacity.

With the Gate-All-Around transition, we are also gaining share, and we're on track to capture over 50% of the process equipment spending for transistor steps.

We also have very strong market share in interconnect, or the wiring used to transmit data at high speed and low power. Our available market for the wiring steps is approximately \$6 billion for each 100,000 wafer starts per month, and we expect it to grow by \$1 billion when backside power delivery is introduced into volume manufacturing.

Overall, we expect to generate more than \$2.5 billion of revenue from Gate-All-Around nodes this year and potentially more than double that in 2025.

In DRAM, one of the key approaches that memory makers are using to improve performance and power consumption is to implement logic technologies in the peripheral circuitry. Our deep capabilities in logic, combined with our strong position in DRAM patterning, and our unique co-optimized hard mask solutions for capacitor scaling, makes us the clear leader in process equipment for DRAM today and best positioned for future growth.

In the critical die stacking technologies used in high-bandwidth memory, we also have strong leadership positions, including in micro-bump and through-silicon via. Last quarter, we said that we expected our HBM packaging revenue to be 4x larger in 2024 than in 2023. As we have recently seen customers accelerate their capacity plans for HBM, we now believe that our revenue could be 6x higher this year, growing to more than \$600 million.

Across all device types, we now expect revenue from our advanced packaging product portfolio to grow to approximately \$1.7 billion this year.

Looking further ahead, we see opportunities for this business to double again, as heterogeneous integration is more widely adopted beyond the AI data center, and we introduce new products that expand our served market.

AI data center is just one example that illustrates how the major inflections that underpin the next-generation of semiconductors are enabled by Applied Materials. Material science and materials engineering are increasingly important to the industry's roadmap. Applied has invested early to develop a broad, unique and connected portfolio of materials engineering solutions that are critical to enable major semiconductor inflections from AI high-performance computing to ICAPS edge computing. We are translating those investments into consistent outperformance.

Recent TechInsights' data confirms that in 2023, Applied grew faster than the wafer fab equipment market for the fifth year in a row. We accomplished this despite headwinds created by trade rules that we estimate restricted us from more than 10% of the China market during that period.

While we have gained share overall, we are growing share within our served market even faster. This is important because as logic devices become more three-dimensional, future generations of DRAM come to market and advanced packaging becomes more prevalent, we expect materials engineering to become an even larger portion of overall wafer fab equipment.

Another key component of Applied's strategy is to address the increasing complexity faced by the industry. First, we are driving earlier, deeper and broader collaboration with customers and partners. We are changing the industry's innovation model with a goal to accelerate mutual success rates and increase investment efficiencies. Our global EPIC platform that we will build out over the next several years is specifically designed to support high-velocity innovation and commercialization of next-generation technologies.

Second, we are able to provide more complete and connected solutions that accelerate major device inflections. The portion of our revenue generated by integrated solutions has grown from approximately 20% in 2019 to 30% today. We expect demand for these fab-in-a-fab type solutions to continue growing, both at the leading-edge and from our ICAPS customers who are serving Specialty IoT, Communications, Auto, Power and Sensor applications.

And third, we are helping customers transfer new technology into high-volume manufacturing faster and then optimize performance, yield, output and cost in their factory operations. This is supporting double-digit growth of our service business. AGS delivered a new record for revenue this quarter, \$1.5 billion, with a significant portion of this coming from subscriptions in the form of long-term service agreements.

Before I hand over to Brice, I will quickly summarize. Applied Materials continues to deliver strong performance in 2024, and we are in a great position to benefit from secular growth trends over the longer-term. Tectonic shifts in technology, including AI, IoT, EVs, and clean energy that are reshaping the global economy are built on semiconductors, driving demand for more chips and new chips with higher performance and better energy efficiency. Key device architecture inflections that underpin the semiconductor roadmap are enabled by Applied Materials. We expect this to support our ongoing outperformance as next-generation chip technologies, including Gate-All-Around logic nodes, high-performance DRAM and HBM, and advanced packaging move into high-volume production.

And finally, the increasingly complex industry roadmap is driving earlier, deeper and broader collaboration between Applied Materials and our customers and partners, accelerating demand for our most advanced, co-optimized and integrated solutions, and supporting double-digit growth in our service business.

Now I'll hand over to Brice.

Brice Hill {BIO 19639921 <GO>}

Thank you, Gary. And I'd like to thank our teams for their focus and execution, which resulted in another strong quarter for revenue and gross margin.

Today, I'll discuss our overall business environment and share insights into our ICAPS business, which serves the IoT, Communications, Auto, Power, and Sensor markets. I'll describe our capital allocation strategy and operating model, demonstrating how we are driving profitable growth and attractive shareholder returns. I'll also summarize our Q2 results and provide our guidance for Q3.

In calendar Q1, the global market for semiconductors grew 15% year-over-year, and we are optimistic that the data center trends Gary outlined will help drive solid growth for the Semiconductor industry.

During the quarter, cloud service providers announced strong capital spending plans, which is good news for our customers. In the market briefing we issued earlier this month, we forecast that the data center market will eventually become the number one driver of leading-edge foundry-logic wafer starts, surpassing PCs and then smartphones in the coming years. Within our business, we had a strong quarter in fiscal Q2 across DRAM, advanced packaging, ICAPS, and services.

In February, we projected that factory utilization would increase across all device types, and it did. Gary discussed how the data center AI megatrend is driving strong demand for our technologies used in leading-edge logic, DRAM, high-bandwidth memory, and other forms of advanced packaging. Our ICAPS business is driven by three additional megatrends, notably IoT and edge computing, electric and autonomous vehicles, and renewable energy.

Across our ICAPS business, one of the largest demand drivers this year is edge computing, especially at the 28-nanometer node needed by smartphone companies and makers of IoT devices for Industrial and Home Automation applications. A second large demand driver is power chips for electric vehicles, where industry leaders are now establishing supply chains to support their long-term growth plans. A third driver is the power chips used to capture renewable solar and wind energy, which are needed to achieve net-zero goals in the decades ahead.

We believe our ICAPS business will remain a large portion of our overall foundry-logic business for several reasons. One, the megatrends we described will continue to increase unit demand for ICAPS chips. Two, we are innovating at the device level, which creates better chips, stimulates new system sales, and extends Applied's position as the highest value partner for our customers. And finally, we are introducing ICAPS products in additional market segments, which will help us broaden our reach and gain share.

Next, I'll summarize our capital allocation strategy and the results it enables. Our efficient business model generates healthy free cash flow. Our first priority is investing in R&D and capital infrastructure to enable profitable growth. And our second priority is growing our dividend per share and using our buyback program to distribute excess free cash flow to shareholders. Specifically, over the past 10 fiscal years, we have reinvested more than \$20 billion in R&D and over \$5 billion in capital additions and distributed more than 90% of free cash flow to shareholders.

Our capital allocation strategy supports our operating model, which I'll summarize. First, we invest over \$3 billion in R&D each year, collaborating closely with our customers to invent new semiconductor technologies that are critical to their competitive positions in

the global megatrends we've described. Second, we design high-volume manufacturing systems that enable our customers to deploy these new chip and advanced packaging technologies at global scale. Third, we manage a global factory and supply chain network to manufacture these systems.

Fourth, we service our systems, which have decades of useful life, helping customers maximize their return on investment by accelerating ramps and optimizing output, yield and cost. Fifth, we reinvest a high percentage of the profits from this activity back into R&D to develop more new technologies and solutions. And finally, we distribute excess cash to shareholders.

An important point is that every tool we manufacture and ship grows our installed base and our service opportunity, which leads to consistent and stable growth for Applied Global Services. In fact, AGS has delivered 19 consecutive quarters of year-over-year growth, spanning two memory downcycles. Over 80% of AGS revenue comes from recurring services and part sales, about 2/3s of which is delivered under long-term service agreements that have a 90% renewal rate.

Connecting this to our capital allocation strategy, AGS has continued to produce more than enough operating profit to fully fund our growing dividend. In March 2023, we announced a 23% increase in our dividend per share. And in March of this year, we announced a 25% increase.

In summary, over the past 10 fiscal years, our operating model has increased company revenue at a compound rate of over 13%, non-GAAP EPS at nearly 30%, free cash flow at 33%, and dividends per share at nearly 12%. Also, over this period, we reduced net shares outstanding by over 30%.

Now I'll summarize our Q2 results. On a year-over-year basis, net sales grew slightly to nearly \$6.65 billion, non-GAAP gross margin grew 70 basis points to 47.5%, non-GAAP OpEx grew 5% to \$1.23 billion, and non-GAAP EPS grew 4.5% to \$2.09.

Turning to segment results. Semiconductor Systems' revenue remained strong at \$4.9 billion and included record ion implant sales. Segment non-GAAP operating margin was 34.9%. Applied Global Services' revenue increased 7% year-over-year to \$1.53 billion and segment non-GAAP operating margin was 28.5%. Our tools under subscription agreement increased by 8% year-over-year and our installed base of chambers surpassed 200,000 for the first time.

Moving to Display, Q2 revenue was \$179 million, and segment non-GAAP operating margin was 2.8%. We are becoming more confident that the OLED technology found in smartphones will be adopted in Notebook, PCs and tablets, whose larger screen sizes would spur an increase in capital investments.

Turning to cash flows in Q2. We generated nearly \$1.4 billion in operating cash flow and nearly \$1.14 billion in free cash flow. We distributed nearly \$1.1 billion to shareholders, including \$266 million in dividends and \$820 million in buybacks. We repurchased 4.1 million shares at an average price of \$197.77.

Now I'll share our guidance for Q3. We expect revenue of \$6.65 billion, plus or minus \$400 million, and non-GAAP EPS of \$2.01, plus or minus \$0.18. Within this outlook, we expect Semi Systems revenue of around \$4.8 billion, AGS revenue of about \$1.57 billion, and Display revenue of around \$245 million. We expect non-GAAP gross margin to be

approximately 47% and non-GAAP operating expenses to be around \$1.26 billion. Finally, we are modeling a tax rate of 12.3%.

Thank you. And now, Mike, let's begin the Q&A.

Michael Sullivan {[BIO 16341622 <GO>](#)}

Thanks, Brice. To help us reach as many people as we can, please ask just one question on today's call. If you have another question, please re-queue and we'll do our best to come back to you later in the session. Operator, let's please begin.

Questions And Answers

Operator

(Question And Answer)

Certainly. And our first question comes from the line of C.J. Muse from Cantor Fitzgerald. Your question, please.

Q - C.J. Muse {[BIO 6507553 <GO>](#)}

Yes. Thank you for taking the question. I guess, I would love for you to talk about your visibility today and how your ongoing conversations are going with your large customers as it pertains, both to your outlook for kind of second half versus first half on a calendar basis, as well as building that potential backlog into calendar '25? Thank you.

A - Brice Hill {[BIO 19639921 <GO>](#)}

Okay. Great. Hi, C.J. Thanks for the question. So, when we look at our business and think about the largest customers, as we look at Q3 and our guidance for Q3, we know that the DRAM shipments that we've had for China that we called out that were fairly high in Q1 and Q2 are falling off in the second half. And what we've been expecting is that ICAPS strength and leading logic strength would backfill that drop off in DRAM, and that's exactly what we see happening in Q3. And that's consistent on the leading logic side with the advent of Gate-All-Around process technology and equipment beginning to ship earnest -- in earnest for the HBM ramp that will be upcoming.

And I think that's a good indication of what we're expecting in the second half. We talked about whether leading logic and ICAPS would be able to fill in for the drop off in China DRAM, and that's exactly what we see in Q3.

And then when we look ahead to '25, will point to the same fundamental ramp for leading logic. That'll be Gate-All-Around. Gary highlighted in his remarks that we expected \$2.5 billion of business related to gate around -- Gate-All-Around shipments in '24, and we expect to be able to grow that significantly in '25 as that process technology ramps towards HVM, high-volume manufacturing.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Stacy Rasgon from Bernstein Research. Your question, please.

Q - Stacy Rasgon {[BIO 16423886 <GO>](#)}

Hi, guys. Thanks for taking my question. Brice, I wanted to -- and Gary, I wanted to follow-up on that sort of qualitative 2025 outlook. So, I guess, I -- I guess, I heard you right. You said \$2.5 billion Gate-All-Around this year doubling next year or more. So, \$2.5 billion of incremental year-over-year growth from Gate-All-Around. It sounds like you said the advanced packaging have the opportunity to double again over time. So, that should be growing. HBM, theoretically, should be growing. ICAPS, I know, has been a little weaker than it was. It feels like it's bottomed just given what we're seeing. That ought to grow. And -- I mean, NAND flash is, for your guys, is pretty close to zero right now as well. So -- and that ought to grow.

I mean, how do we think about, like, you've talked about over-delivering versus the market. Like, let's say, the WFE market was flat next year, given those drivers, how would I think about how you guys could grow relative to a market like that, given those drivers?

A - Brice Hill {BIO 19639921 <GO>}

Yes. Thanks, Stacy, for the question. So, the way our perspective, we start with the longer-term view. And as Gary described, we're investing in the key inflections for the fastest-growing markets. And so, our perspective, if you stand back and think over many quarters or the next several years, is that the leading logic technologies like Gate-All-Around and backside power, the advanced packaging, HBM memory, our services business, we think all of those will grow faster than the average market and will help us gain share. So, it's harder to call any particular quarter, or short time period in the near future, but that would be our perspective.

And I think from ICAPS perspective, you called out Gate-All-Around and ICAPS and DRAM and NAND, I think -- thinking about all of them, we've talked about our DRAM share having improved over 10 percentage points over the last 10 years, and we expect that technology has great tailwinds. We talked about how HBM and utilization in DRAM is improving, so that's a positive.

And if we think about ICAPS, ICAPS, the market, we have three quarters here that you can see, two -- our first two that are closed and our Q3 guidance, and ICAPS is very strong in those quarters and grows in Q3 to fill in some of that China DRAM business that drops off.

And then finally, Gate-All-Around, the \$2.5 billion that Gary highlighted, that's not just the incremental. That's our shipments for the entire business that are associated with Gate-All-Around that are shipping this year, and we do expect that to grow next year.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Stacy, this is Gary. Really, the big focus for us is enabling and winning major device architecture inflections. So, when you break out all of these different device segments, certainly foundry-logic leading-edge, the opportunity there for us to grow is significant because we have a very high share of those big inflections, the Gate-All-Around, backside power delivery, wiring is a real strength for us. As Brice mentioned, in DRAM, we gained 10 points of overall DRAM share over the last 10 years.

And if I look forward at the architecture inflections in DRAM, we're even better positioned. Those will be much more materials-intensive, materials engineering-enabled, and that's going to be a really good position for us.

We also talked about packaging. We're -- we've increased our view of packaging in '24 to \$1.7 billion. But as I mentioned and Brice reiterated, we have an opportunity to double advanced packaging -- our advanced packaging business. And HBM, that's -- we've increased that growth from 4x to 6x this year.

And the last one is our services business, 19 consecutive quarters year-over-year where we've seen growth. And with these highly-complex architecture inflections, it gives us a really great opportunity to add more value with our services business. Also, as people are moving into new locations where they don't have an experienced workforce, we've talked about double-digit growth in services. Right now, that's a \$6 billion run rate, but that's adding significant growth as we go forward. So, those are just a few other thoughts.

Q - Stacy Rasgon {BIO 16423886 <GO>}

But you said \$2.5 billion this year from Gate-All-Around and more than double that in 20 - did I hear that wrong? Does that suggest \$5 billion in Gate-All-Around in 2025?

A - Gary E. Dickerson {BIO 2135669 <GO>}

Yes, that's what --

Q - Stacy Rasgon {BIO 16423886 <GO>}

Or did I misunderstand what you said?

A - Gary E. Dickerson {BIO 2135669 <GO>}

No, I think that math is correct.

A - Brice Hill {BIO 19639921 <GO>}

Yes. That's right, Stacy. So, that's not just the increment. That's the entire equipment sales that we'll have for that technology. So, \$2.5 billion for this year and opportunity to double that next year.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Got it. Thank you, guys.

A - Brice Hill {BIO 19639921 <GO>}

Yes.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Mark Lipacis from Evercore ISI. Your question, please.

Q - Mark Lipacis {BIO 2380059 <GO>}

Hi. Thanks for taking my question. Maybe a question for Gary. Gary, when you talk about like investing more in materials engineering, it seems like you're kind of climbing up the value chain to the place where your -- it used to be your customers' domain. And a lot of times when you see companies moving up the value chain and adding more value, ultimately you see that reflected in improving margins.

So, appreciate that you engage in partnerships and that's the spirit. But like if you're adding more value, it seems like that ultimately at the end of the day should come back to you in higher profitability. And I'm wondering if you could just talk philosophically about that. Like how -- is that something that you would expect to see happen or does that manifest in other ways? Thank you.

A - Gary E. Dickerson {[BIO 2135669 <GO>](#)}

Yes. Thanks for the question, Mark. So, if you look at the industry overall, all our customers are racing to be first-to-market to deliver those major device architecture inflections that determine their competitive position. So, Gate-All-Around, backside power, those new DRAM architectures and packaging is becoming more and more important to power performance and cost. So, that -- that is really what shapes the competitive landscape in the industry. And it's very, very clear. We're working with our customers for technology generations out into the future. So, we have very high visibility and this broad and connected portfolio that we have, gives us an opportunity to engage with customers in those early and very deep relationships on those architecture inflections.

And as we look forward, we do see materials engineering as a percentage of spend in the relative contribution increasing. And Applied has clear leadership, especially with this broad connected portfolio. We've built integration capability inside Applied that is really tremendous and really co-innovating with our customers as they're driving those architecture inflections.

So, I do believe for sure that we will be creating more value in those relationships with our customers, and that certainly is valuable for them. And our focus is to capture more value as we're delivering more value with them. So, your basic question, I agree with your thoughts.

Q - Mark Lipacis {[BIO 2380059 <GO>](#)}

Thanks. Very helpful.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Vivek Arya from Bank of America Securities. Your question, please.

Q - Vivek Arya {[BIO 6781604 <GO>](#)}

Thanks for taking my question. Gary, maybe back to the Gate-All-Around. So, you're raising the forecast there by \$1 billion. I think on, I think, HBM, you're raising it by another \$100 million or so. Is that all going to show up in July and October? So, does it mean that conceptually, the second half of the year is over \$1 billion higher than what you thought before?

And then on Gate-All-Around, what is driving such a significant growth versus what you thought before? \$1 billion increase over \$1.5 billion base is a very strong number. So, I'm just curious what are the kind of the qualitative factors? And then just also quantitatively, does it mean that second half is \$1 billion higher -- \$1.1 billion higher than what you thought before?

A - Brice Hill {BIO 19639921 <GO>}

Hi, Vivek. It's Brice. I'll just do a clarification here. This quarter -- we decided last quarter when we talked about the incremental revenue for Gate-All-Around that it was a little bit confusing. So, we changed and this quarter, we're talking about all of our revenue that's associated with the Gate-All-Around transistor and wiring. And so, that's not just the increment. So, the \$2.5 billion that we're talking about this quarter is not an -- is not supposed to be interpreted as an increase relative to the \$1 billion we talked about last quarter. We're just changing the basis. Now we're just quoting all of the business that's associated with the Gate-All-Around product. So, hopefully that helps.

So, as far as answering the question about what's driving the increase, I don't think our perspective has changed very much. It's \$2.5 billion for this year. It can grow significantly next year as the technology ramps. And then what I would say is, you do see some of that in our Q3 guide. So, as you know, we took down -- or we're dropping off the incremental China DRAM shipments that we had. And in the second half, we're filling that in with ICAPS strength and growing leading logic shipments that are driven by Gate-All-Around.

Q - Vivek Arya {BIO 6781604 <GO>}

But just a clarification --

A - Gary E. Dickerson {BIO 2135669 <GO>}

Vivek, I'd add that relative to kind of what drives the incremental spending, these technologies where you're improving device performance, the drive current, leakage, power consumption, all of those things are incredibly difficult. And so, the customers -- and it's really all about this materials magic that we're able to provide with that broad portfolio of different types of technologies. So, step counts are increasing at every one of those different technology nodes as they are driving those roadmaps for performance, power to meet the -- especially for high-performance computing, AI computing demand type of applications.

So, again, those are very, very, very, very difficult technologies, whether it's 100 billion transistors in a GPU or 100 kilometers of wiring in an applications processor. Those are really amazing, amazing technology accomplishments with materials. That's what drives that incremental spending.

Q - Vivek Arya {BIO 6781604 <GO>}

Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Chris Caso from Wolfe Research. Your question, please.

Q - Chris Caso {BIO 4815032 <GO>}

Yes. Thank you. Good evening. The question is on advanced packaging, and you were pretty specific with what you're expecting this year. You spoke about the opportunity to double that again. You didn't provide a timeframe on that. So, if you could help us with some of the timeframe on when you expect to double that?

And just in terms of what's driving that, we know there's advanced packaging in logic and HBM. What's the mix of that in terms of what's driving the expectations higher?

A - Gary E. Dickerson {[BIO 2135669](#) <GO>}

Hi, Chris. This is Gary. So, this whole focus around advanced packaging, heterogeneous integration, is really a major drive across the entire industry. When you look at AI servers or any of these different kinds of applications, you see tremendous innovation. In the high-bandwidth memory, you see tremendous innovation and how you're connecting together all of those high-performance logic chips and memory chips.

And we really see the growth there being very, very significant because it's so important to the industry's roadmap, as kind of classics Moore's Law has slowed down significantly, this is another major driver of how the industry will innovate going forward.

And with Applied, we have the broadest portfolio of technologies, PVD, CVD, CMP, plating, Etch. We've added hybrid bonding, digital lithography, eBeam test. We have a very broad portfolio. And as we see those roadmaps, and this is where I come back to architecture inflections are really what drives your incremental business going forward. So, we can see the roadmaps for our foundry-logic customers, for memory customers, all of them are investing very heavily in these types of technologies. And some of these areas also that are new innovations from Applied will help us drive that business 2x higher. And I think the compound annual growth rate keeps going up from there.

The other advantage that we have besides this broad portfolio and the deep multi-generation connection that we have with all of our customers is our full-flow packaging lab in Singapore. We're working with leading customers on those next-generation innovations, Chris.

So, we have a pretty good line of sight to what those inflections look like, the architecture inflections will look like in a very strong position to grow our share as those inflections come to market. We're not giving a specific timeframe on the doubling. It will happen over a number of years, but very, very optimistic about our opportunities in packaging.

Q - Chris Caso {[BIO 4815032](#) <GO>}

Thanks.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Srini Pajjuri from Raymond James. Your question, please.

Q - Srini Pajjuri {[BIO 21216060](#) <GO>}

Thank you. Brice, I have a question on China. I think last quarter, you said you expect China to normalize to about 30% or so by the end of this year. So, my question is, is that still the expectation? And then it looks like ICAPS is still holding up pretty well. So, is the decline primarily DRAM going forward? And I'm just curious as to given what we are hearing about what we're seeing in the end markets of ICAPS, I mean, it looks like the broader analog is still relatively weak and auto is kind of mixed. I'm just curious as to what's driving the strength in ICAPS? Thank you.

A - Brice Hill {[BIO 19639921 <GO>](#)}

Okay. Thanks, Srini. Yes, first of all, our mix as we move to the second half of the year will normalize with respect to China as a percent of our total revenue. So, it'll be closer to 30% as expected, and it's the dynamic that you highlighted. We had some catch-up DRAM shipments to particular customers in China for the last three quarters. Those will fall off as we go through Q3 and Q4, and that really will bring our percentage of China revenue back down to closer to what's averaged over the last few years, which, we'll call about 30%.

And then with respect to the mix of business, so as that DRAM business falls off, our ICAPS business and our leading-edge logic business does increase to fill that in. And so, that's consistent with continued strength in ICAPS, which, the rest of the market for us in China would be ICAPS-related.

And I think you described it well. The end markets are mixed. So, industrial and auto have been weaker. Smartphone, PC-related products have been slow, but gaining some strength recently. And then as we've talked about image sensors, power chips, microcontrollers, a lot of those markets have been stronger. So, it's been mixed end markets, but the customers are investing for forward-looking demand and to get capacity in place for forward-looking demand. And when we see utilization in those markets, we've seen it improve.

So, I think when people ask us for guidance, hopefully the best thing we can tell you is Q1 and Q2 are very strong in ICAPS and Q3 will be another strong quarter in the ICAPS market. Thanks for the question.

Q - Srini Pajjuri {[BIO 21216060 <GO>](#)}

Thanks, Brice.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Atif Malik from Citi. Your question, please.

Q - Atif Malik {[BIO 7312618 <GO>](#)}

Thank you for taking my question. I have a question on Gate-All-Around as well. Very strong outlook, \$2.5 billion doubling to next year. Is the customer funnel for this demand pretty broad across the core foundries? And also, if you can rank order for us in the \$2.5 billion Epi, ALD, patterning, like what is -- what's driving the most demand? Thank you.

A - Brice Hill {[BIO 19639921 <GO>](#)}

I think I'll take the first part. I think, that demand is across our customer base. So, it's not one single customer, but we won't share any details about the relative balance between customers.

And then from a device mix perspective, Gary -- or I'm sorry, on equipment mix, I don't know if there's anything you would highlight there.

A - Gary E. Dickerson {[BIO 2135669 <GO>](#)}

Yes. I think, again, one of the advantages that we have is this broad portfolio of technologies, along with integrated tools and also our integration engineering, where we're working with customers and really co-innovating on these different technology inflections. So, if you look at Gate-All-Around, we have Epi, PVD, ALD, Selective removal, Etch, Thermal processing, Implant, CMP, and eBeam. So, it's a very broad portfolio.

And I'm not going to break out all of those different pieces, but what I would say is our ability to connect that portfolio together is really a tremendous strength in our ability to co-optimize as you're thinking about, and we've done this back in FinFET also, where we were able to co-optimize and innovate in how you're building those structures so that you can improve power and performance, and we're doing the same thing with Gate-All-Around.

Operator

Does that answer your question?

Q - Atif Malik {BIO 7312618 <GO>}

Yes. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Toshiya Hari from Goldman Sachs. Your question, please.

Q - Toshiya Hari {BIO 6770302 <GO>}

Hi. Good afternoon. Thank you so much for taking the question. I had two quick ones, one on HBM and the other one on NAND.

So, in HBM, your customers would tell us that they're sold out certainly for '24. For the most part, they're sold out for '25 as well. So, my guess is you've got pretty good visibility into next year. Gary, you talked about overall advanced packaging doubling over the medium- to long-term. But when you zoom into HBM, could that business for you multiply again in '25 on a year-over-year basis?

And then on the NAND side, the business continues to be pretty depressed. Your customers, their profitability is improving, cash flow is improving or becoming less bad. Are you starting to see purchase orders from your customers come back, or what do they need to see for them to spend on WFE again? Thank you.

A - Brice Hill {BIO 19639921 <GO>}

Okay. Toshiya, it's Brice. I'll start and then Gary may add something on the HBM. So, on the sold-out comment, what we do -- we have seen utilizations improve across the DRAM wafer starts. We've seen the allocation to HBM from a wafer start perspective increase from 5% to probably something closer to 20%. That said, it's not 100% utilized. So, you can speculate what the -- if there's some restraint. It could be in the packaging side, and we have seen orders for HBM going up. So, we'll have to let you think about that and talk to those vendors from that perspective.

On the NAND -- on the NAND side, our perspective on NAND is that Moore's Law is still very much alive in NAND. The bit density is increasing with each generation of NAND fairly

aggressively, and it's meeting the demand function for increased bits. And what that really means is that the business we see in NAND is for technology upgrades, more technology upgrades, not really new wafer starts. So, we -- that's what we expect to see as we move forward. And overall, we expect NAND to grow at the speed of semiconductors. It's -- obviously, storage has a very important part, and several of the vendors called out how important storage is, even for AI, and that it will grow. So, I think for us, it's been slow. But we expect it to grow over time, and memory in total to be about 1/3 of our sales for WFE.

And then, Gary, back to you, whether anything specific on HBM.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Yes. Really -- hi, Toshiya. Nothing really too much more beyond what Brice was talking about. And we have seen the demand for HBM strengthening. We went from about 4x increase to 6x increase from the last time that we talked to you guys. And I would say that we still are seeing incremental demand going forward. So, that pull is still there from customers. We're not giving any point estimate for next year, but we do continue to see stronger demand in that segment.

Q - Toshiya Hari {BIO 6770302 <GO>}

Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Krish Sankar from TD Cowen. Your question, please.

Q - Krish Sankar {BIO 16151788 <GO>}

Yes. Hi. Thanks for taking my question. Gary, I had a question on backside power delivery. And it seems that the industry has two approaches to it, one is the backside wiring with direct contact to the transistors and another one is the nano-TSV approach. I'm kind of curious, you've spoken about the 100 -- incremental \$1 billion SAM for 100,000 wafer starts a month. How does this split between those two approaches? And also, can you quantify what kind of revenues you expect this year and next year for Applied in backside power delivery? Thank you.

A - Gary E. Dickerson {BIO 2135669 <GO>}

Hi, Krish. Yes, I don't want to talk too much about the specific architectures that customers are using. It's confidential for all of those different customers. I would say that going forward, that direct connect to the backside is the path that everybody is focused on. So, I think over time that that will grow a fairly significant amount. Well, both of those approaches will grow a fairly significant amount.

And we're deeply engaged with customers on that inflection. We're leading in wiring overall. We have a very, very high share of wiring. And so, wiring going to the backside, we've said that that gives us an opportunity of 50% of that spend when that inflection happens.

So, we're supporting both of those different types of approaches for the backside. But I don't want to get too specific in terms of which one do we see bigger. Over time, I think

that it could move to the direct connect. But that timing, I don't want to give specifically because that's going to be each one of the different customers' roadmaps. Relative to timing for revenue, it's pretty small for us right now in 2024. It will grow in '25, but the ramp -- significant ramp for backside power is still out beyond '25 in revenue.

Q - Krish Sankar {BIO 16151788 <GO>}

Thanks, Gary. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Timothy Arcuri from UBS Securities. Your question, please.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Thanks a lot. Brice, I had a two-part question on China DRAM. And the first part is, you talked about it coming down to \$500 million. Is that stepping all the way down in fiscal Q3? So, what's the assumed step down in China DRAM for fiscal Q3?

And then -- and then, more broadly, of the \$1.6 billion worth of DRAM business, I'm wondering if you can kind of help us figure out how much of that is China? If I look at Korea and I look at how much that is, it looks like about \$1 billion of the \$1.6 billion DRAM revenue is China. Can you confirm whether that's in the right ballpark? Thanks.

A - Brice Hill {BIO 19639921 <GO>}

Hi, Tim. Thanks for the question. The only quantification that I gave in prior quarters was that our DRAM as a whole was up more than \$500 million in the quarters where we had that incremental. So, I think I'll just leave it at that and say it's more than \$500 million in each of the last three quarters. And as we look into the second half, it doesn't drop completely to zero in Q3, but it drops significantly, and then it's pretty close to zero in Q4. In our mix to China revenue for the company, as I highlighted to an earlier question, will be about in the range of 30%, which is normal for us. So, I think that's as close as I can get on that. Thanks for the question.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Sure, Brice. Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Joe Quatrochi -- Quatrochi from Wells Fargo. Your question, please.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Yes. Thanks for taking the question. I wanted to kind of try to understand just the trends you're seeing in ICAPS. Can you help us just understand the difference in the growth that you're seeing from China versus non-China? Because it sounds like, as we look at China revenue normalizing, the DRAM falling off, obviously, the rest of that's mostly just ICAPS. So, can you help us just kind of understand the trends between China and non-China?

A - Brice Hill {BIO 19639921 <GO>}

Hi, Joe. Thanks for the question. Yes, for China, what I would say is the last three quarters have been very strong. So, if we're speculating about a slowdown or a digestion period, one of the reasons I didn't reemphasize that is we're kind of going by what we're seeing. So, we had two strong quarters, Q1, Q2. Our Q3 guide, again, has strength in ICAPS and a strong quarter for China. When we think about the world right now, I think we have three regions growing and the rest of the regions not growing. So, that corresponds with the end markets sort of being mixed at this point.

Our view of China overall, the team showed a pie chart of the different device elements of WFE last year. ICAPS was about 40% of WFE by that pie chart on May 2 that was put out. China is our largest market in ICAPS, and we monitor that carefully. We've got about -- we've got a large number of customers. We're tracking about 30 different -- more than 30 factory projects that are in ramp process. In other words, they have capacity, they're adding capacity, they're ramping. We're tracking utilizations. Utilizations are improving. We understand yields are improving. So, we think that the China market will be a very large and important market for us going forward. We said we don't see a cliff for that market, and I think we'd stick by all those remarks at this point.

Q - Joe Quatrochi {BIO 18961101 <GO>}

Thank you.

Operator

Thank you. One moment for our next question. And our next question comes from the line of Harlan Sur from JP Morgan. Your question, please.

Q - Harlan Sur {BIO 6539622 <GO>}

Good afternoon. Thanks for taking my question. As we transition from a more mature specialty ICAPS-driven growth profile to a leading-edge sort of advanced foundry-logic memory spending profile this year and going forward, this typically tends to be a strong tailwind for your process control and diagnostics portfolio, which leans very heavily on your most advanced tools and solutions, right, because feature sizes get smaller, so better resolution required, but also new and more defect and yield challenges, right? So, it's probably a driver of your leading-edge CFE-driven eBeam portfolio this year. Are you guys still seeing that portfolio grow like 4x this year? And how does the pipeline look for next year?

And maybe mid- to longer-term, like given the significant manufacturability challenges, does the Applied team see process control and diagnostics intensity rising faster than overall WFE intensity over the next few years?

A - Gary E. Dickerson {BIO 2135669 <GO>}

Thanks, Harlan, this is Gary. So, I would say that, for sure, the connection that we have with our eBeam technology and these major architecture inflections is a real advantage for us because, again, it's a race on who gets there first relative to those major architecture inflections and learning rate, how fast you learn is one of the determining factors. So, having this unique capability is a real advantage for Applied. I mean, obviously, it's a good growth driver for us from a revenue perspective, but that synergy with the rest of our broad portfolio, our connected portfolio --

Q - Harlan Sur {BIO 6539622 <GO>}

Yes.

A - Gary E. Dickerson {BIO 2135669 <GO>}

-- is a real advantage. On cold field emission, which is really one of the key technologies in the electron optics, it gives us the highest resolution, add up to 10x higher imaging speeds. So, it really is highly differentiated, and we are still on track this year to grow our CFE systems' revenue around 4x in '24, and that would represent about 50% of our total eBeam system sales.

And then when I look forward, again, I see PDC as a really great growth driver for Applied. We've had the strongest pipeline of new products than we've ever had and the synergy with the rest of our process equipment business, so all of that materials' -- magic materials innovation is increasing. So, again, very, very optimistic about the shape of that business.

Q - Harlan Sur {BIO 6539622 <GO>}

Thank you, Gary.

A - Michael Sullivan {BIO 16341622 <GO>}

Yes. Thanks, Harlan. And operator, we have time for just one more quick question, please.

Operator

Certainly. One moment for our final question. And our final question for today comes from the line of Mehdi Hosseini from Susquehanna. Your question, please.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

Yes. Thanks for letting me ask a question. Gary, you were talking about this inflection point technology, leading-edge logic, HBM, DRAM, advanced packaging. But would it be fair to say that what is not in your backlog is the additional wafer capacity, especially for Gate-All-Around and memory, and that would be something that would be coming in the POs be placed later this year for shipment in 2025?

A - Brice Hill {BIO 19639921 <GO>}

Yes. Hi, Mehdi. It's Brice. I guess, I'll take that one. We do expect -- I can't tell if it's a greenfield question or not, but we do expect increased wafer starts across definitely ICAPS, definitely leading-logic, a smaller amount in DRAM, and probably the place where we don't really expect increased wafer starts is in NAND, but that will still be upgrades. So, most of the memory technologies are upgrades, but it'll be a combination of new technologies and new plant and equipment across the board.

Q - Mehdi Hosseini {BIO 4362002 <GO>}

Thank you.

A - Michael Sullivan {BIO 16341622 <GO>}

Okay. Thanks, Mehdi, for your question. And now, Brice, would you like to give us your closing thoughts.

A - Brice Hill {[BIO 19639921](#) <GO>}

Thanks, Mike. I think we've done a good job anticipating the roadmap inflections in data center AI. We can see the pull for our solutions in Gate-All-Around chips from transistors to frontside wiring and backside wiring and in advanced packaging. Same goes for DRAM, where we're number one in materials engineering and especially strong in HBM stacking. In future calls, we'll have a lot more to say about our positions in edge, AI and IoT, plus automotive and clean energy, which are very big and long-term drivers of our ICAPS business.

The momentum in our systems business fuels our service business, which is driving very stable subscription-like growth and helping us increase the dividend at an accelerated rate. Gary will be at the Bernstein Conference in New York on May 30, and I hope to see many of you at the BofA Conference in San Francisco on June 6.

Thank you. Mike, let's close the call.

A - Michael Sullivan {[BIO 16341622](#) <GO>}

Okay. Thanks, Brice. And we'd like to thank everybody for joining us today. A replay of today's call is going to be available on the IR page of our website by 5 P.M. Pacific Time today. And we'd like to thank you for your continued interest in Applied Materials.

Operator

Thank you, ladies and gentlemen, for your participation in today's conference. This does conclude the program. You may now disconnect. Good day.

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